Chapter 11: Conic Sections

Name: ______________________ Date: ______________________

Instructions: Answer all questions and show clear work. Give exact values when possible. For each conic, include key features (vertex/center, foci, directrix, asymptotes) as requested and provide graphs when instructed.

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1. **Parabolas.** (10)

   (a) Find the vertex, focus, directrix, and axis of symmetry of $(y+1)^2 = 12(x-2)$. Sketch the graph.

   (b) Find the equation of the parabola with vertex $(-2, 3)$ whose directrix is $x = -6$. State the focus and orientation.

2. **Ellipses.** (12)

(a) For $\frac{(x-1)^2}{36} + \frac{(y+2)^2}{9} = 1$, find the center, vertices, co-vertices, foci, and lengths of the axes.

(b) An ellipse has foci at $(\pm 10, 0)$ and passes through $(0, 9)$. Find its equation in standard form and its eccentricity.

3. **Hyperbolas.** (12)

   (a) For $\dfrac{(x+3)^2}{4} - \dfrac{(y-1)^2}{25} = 1$, find the center, vertices, foci, and equations of the asymptotes.

   (b) A hyperbola centered at $(0,0)$ has asymptotes $y = \pm\frac{3}{4}x$ and a vertex at $(0,6)$. Find its equation.

4. Classify each conic, convert to standard form by completing the square, and list the requested features. (12)

   (a) $x^2 + y^2 + 8x - 6y + 9 = 0$ (center and radius)

   (b) $9x^2 - 4y^2 - 54x - 16y - 61 = 0$ (center, vertices, asymptotes)

5. **Rotation of Axes.** (12)

   (a) Find the angle that eliminates the xy-term in $x^2 - \sqrt{3}\,xy - y^2 = 4$. Write the rotated equation in u, v coordinates and identify the conic.

   (b) The graph of $xy = 1$ is rotated 45° counterclockwise. Write the equation in standard form and list the asymptotes in the original x, y axes.

6. **Polar Conics.** (12)

   (a) A conic with focus at the origin has eccentricity $e = 1$ and directrix $x = 2$. Find its polar equation and state its type.

   (b) Determine the eccentricity and a suitable directrix for $r = \dfrac{8}{3 - 2\sin\theta}$. Classify the conic.

7. **Mixed Skills.** (10)

(a) For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $a > b > 0$, show that the latus rectum length (through a focus, parallel to the minor axis) equals $\frac{2b^2}{a}$. Give a brief derivation.

(b) Prove that the rectangular hyperbola $xy = c^2$ has asymptotes $x = 0$ and $y = 0$ and eccentricity $e = \sqrt{2}$ after rotation to standard position.

8. **Focus on Modeling: Headlight and Architecture.** (20)

   (a) A car headlight uses a parabolic reflector. The bulb (focus) is 3 cm in front of the vertex. The rim of the reflector is 20 cm from the vertex along the axis. What is the diameter of the opening? Provide the parabola equation and your computation.

   (b) An arch is modeled by a parabola opening downward with vertex at height 10 m and spanning 24 m at the ground. How high is the arch 3 m from one side? Then compare with an elliptical model having the same span and a central height of 10 m; compute the height at the same location. Which model gives more clearance near the sides?